

Exercise Sheet 7

Uncertainty Quantification

Calibration Concepts

Exercise 7.1: Epistemic vs. Aleatoric Uncertainty

optional

Define *epistemic* and *aleatoric* uncertainty.

1. Which type is more relevant to OOD inputs (e.g., night images for a model trained on day images)?
2. Which type would you expect to be dominant in a correctly classified, in-distribution image?

Exercise 7.2: Calibration and ECE

Explain what it means for a classifier to be *well-calibrated*. Define the Expected Calibration Error (ECE) and describe how it is computed.

Exercise 7.3: Cost-Optimal Downstream Decisions

A pedestrian detector outputs $p = p(\text{ped} \mid x) \in [0, 1]$. The autopilot must choose between two actions with asymmetric costs:

	pedestrian present	no pedestrian
BRAKE	0	$C_{\text{FP}} = 1$
CONTINUE	$C_{\text{FN}} = 100$	0

At which p should the autopilot brake in order to minimize the costs?

1. Write down the expected loss (costs) of each action as a function of p : $\mathbb{E}[\mathcal{L} \mid \text{BRAKE}]$ and $\mathbb{E}[\mathcal{L} \mid \text{CONTINUE}]$.
2. Find the threshold τ^* at which both actions have equal expected loss. For $p > \tau^*$ the autopilot should brake; for $p < \tau^*$ it should continue.
3. Compute τ^* for the costs above. How does it compare to the standard argmax rule ($\tau = 0.5$)?
4. Why does τ^* only yield the optimal decision when p is well-calibrated? What can go wrong when the same threshold is applied to an overconfident model?

Practical: Calibrating the CARLA Model

Exercise 7.4: Measuring Calibration

1. Compute the ECE of each of your three trained CARLA models on the in-distribution test set.
2. Plot a *reliability diagram* (confidence vs. accuracy) for each model.
3. Are your models over- or underconfident? Does this pattern hold consistently across all three models?

Exercise 7.5: Temperature Scaling

Temperature scaling adjusts the model's logits as:

$$p(y | x) = \text{softmax}\left(\frac{f(x)}{T}\right) \quad (1)$$

1. Apply temperature scaling to each of your three models. Optimize T on the validation set using negative log-likelihood.
2. Report the ECE before and after scaling for each model.

Hint: A simple line search is sufficient: evaluate the validation NLL for a grid of values such as $T \in \{0.5, 0.6, 0.7, \dots, 3.0\}$ (step 0.1) and pick the T with the lowest NLL.

Exercise 7.6: Cost-Optimal Decision in Practice

Use the costs from Exercise 3 ($C_{\text{FN}} = 100$, $C_{\text{FP}} = 1$, $\tau^* \approx 0.0099$, i.e. a predicted probability of roughly 1 %) and apply them to your *pedestrian* classifier.

1. On the in-distribution test set, compute the total loss

$$\mathcal{L} = C_{\text{FN}} \cdot \#\text{FN} + C_{\text{FP}} \cdot \#\text{FP}$$

for your **uncalibrated** model at both $\tau = 0.5$ and $\tau = \tau^*$.

2. Repeat for your **temperature-scaled** model at both thresholds.
3. Arrange the four results in a 2×2 table (rows: uncalibrated / calibrated; columns: $\tau = 0.5$ / $\tau = \tau^*$). Which combination gives the lowest total loss?

Exercise 7.7: Tracing Overconfidence Through the Safety Analysis

You have now measured how miscalibrated confidence propagates silently to downstream decisions. In STPA, an overconfident misclassification is a **causal factor**: a mechanism that explains *why* an existing unsafe control action occurs. The goal of this exercise is to add it as a loss scenario and trace it through to a verifiable constraint.

1. **Causal scenario.** Revisit your causal loss scenarios from [Exercise 2.8](#). The hazard (“vehicle fails to brake for a pedestrian”, [Exercise 2.4](#)) and the unsafe control action (“the planner does not command braking while a pedestrian is in the path”, [Exercise 2.6](#)) already exist - you do *not* need new ones. Add a causal scenario describing a new way that UCA arises: the pedestrian classifier produces a false negative *yet reports high confidence*, so the planner registers “no pedestrian, reliable” and no low-confidence fallback is triggered.
2. **Safety constraint.** From the scenario, derive at least one new safety constraint of each kind (cf. [Exercise 2.7](#)):
 - a **model-level** constraint on calibration, expressed with the ECE you measured; and
 - a **system-level** constraint on how the planner uses the confidence score - e.g. the fallback when confidence is low, or adopting the cost-optimal threshold τ^* from [Exercise 3](#) instead of 0.5.

3. **Verification.** What evidence verifies the model-level constraint? You produced exactly this evidence on this sheet (the ECE before and after temperature scaling, Exercise 7.4–Exercise 7.5). Does your calibrated model meet the threshold you set?
4. **Residual risk.** Even with a perfectly calibrated model, what does the constraint *not* cover? Does calibration alone close the scenario, or is the system-level fallback still required?

q3. Verification:

The model-level constraint is verified using the evidence: which were ECE measurements before and after temperature scaling obtained in Exercises 7.4 and 7.5.

for my ped model,

- ECE before calibration: 0.617

- ECE after temperature scaling: 0.427

If the safety constraint is $ECE < 0.25$, then:

The calibrated pedestrian model does not satisfy the constraint, since its ECE after calibration (0.427) is still above the required threshold.

q4. Residual risk:

Calibration alone does not eliminate false negatives (C_{fn}) or guarantee safe behavior.

Even a perfectly calibrated model may still miss pedestrians.

Therefore, calibration alone does not close the scenario, and the system-level fallback such as minimal-risk behavior or the cost-optimal threshold *) is still required.

For example, when the model reports low confidence or uncertain predictions, the planner should transition to a minimal-risk state by slowing down, stopping, or requesting human intervention. Additionally, using the cost-optimal threshold * instead of the default threshold i.e 0.5, reduces the risk of missed pedestrian detections and improves overall safety.